Complex Analysis Homework 1

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7. The family of mappings introduced here plays an important role in complex analysis. These mappings, sometimes called Blaschke factors, will reappear in various applications in later chapters.

(a) Let z, w be two complex numbers such that $\overline{z}w \neq 1$. Prove that

$$\left|\frac{w-z}{1-\overline{w}z}\right| < 1 \quad \text{if } |z| < 1 \text{ and } |w| < 1,$$

and also that

$$\left|\frac{w-z}{1-\overline{w}z}\right| = 1 \quad \text{if } |z| = 1 \text{ or } |w| = 1.$$

[Hint: Why can one assume that z is real? It then suffices to prove that

$$(r-w)(r-\overline{w}) \leq (1-rw)(1-r\overline{w})$$

with equality for appropriate r and $|w|$.]

(b) Prove that for a fixed w in the unit disc $\mathbb{D}$, the mapping

$$F : z \mapsto \frac{w-z}{1-\overline{w}z}$$

satisfies the following conditions:

(i) F maps the unit disc to itself (that is, $F : \mathbb{D} \to \mathbb{D}$), and is holomorphic.

(ii) F interchanges 0 and w, namely $F(0) = w$ and $F(w) = 0$.

(iii) $|F(z)| = 1$ if $|z| = 1$.

(iv) $F : \mathbb{D} \to \mathbb{D}$ is bijective. *[Hint: Calculate $F \circ F$.]*

证明 *(a)* 考虑 $|\frac{w-z}{1-\overline{w}z}|^2 = \frac{(w-z)(\overline{w}-\overline{z})}{(1-\overline{w}z)(1-w\overline{z})}$

故只需要证明

$$(w-z)(\overline{w}-\overline{z}) < (1-\overline{w}z)(1-w\overline{z})$$

化简得

$$w\overline{w} + z\overline{z} < 1 + w\overline{w}z\overline{z}$$

，即

$$(1-|w|^2)(1-|z|^2)) > 0$$

根据 $|z|<1$ *and* $|w|<1$，$(1-|w|^2)(1-|z|^2)) > 0$ 成立

第二个式子同理

(b)

(i) 由 *a* 即可得*"F maps the unit disc to itself"*，由于 $w-z$ 和 $1-\overline{w}z$ 都是纯的且 $1-\overline{w}z \neq 0$，于是 F 是纯的

(ii) 代入计算即得

(iii) 根据 *a* 即得

(iv) 在 $\mathbb{D}$ 上计算有 $F \circ F = id$，因此 F 是双射

8. Suppose U and V are open sets in the complex plane. Prove that if $f : U \to V$ and $g : V \to \mathbb{C}$ are two functions that are differentiable (in the real sense, that is, as functions of the two real variables x and y), and $h = g \circ f$, then

$$\frac{\partial h}{\partial z} = \frac{\partial g}{\partial z}\frac{\partial f}{\partial z} + \frac{\partial g}{\partial \overline{z}}\frac{\partial \overline{f}}{\partial z}$$

and

$$\frac{\partial h}{\partial \overline{z}} = \frac{\partial g}{\partial z}\frac{\partial f}{\partial \overline{z}} + \frac{\partial g}{\partial \overline{z}}\frac{\partial \overline{f}}{\partial \overline{z}}.$$

This is the complex version of the chain rule.

证明 设 $f = u + iv, g = g(v,u)$, 从而

$$\frac{\partial h}{\partial x} = \frac{\partial g}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial g}{\partial v}\frac{\partial v}{\partial x}$$

$$\frac{\partial h}{\partial y} = \frac{\partial g}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial g}{\partial v}\frac{\partial v}{\partial y}$$

于是

$$\begin{aligned}\frac{\partial h}{\partial z} &= \frac{1}{2}(\frac{\partial h}{\partial x} + \frac{1}{i}\frac{\partial h}{\partial y}) \\ &= \frac{1}{2}(\frac{\partial g}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial g}{\partial v}\frac{\partial v}{\partial x} + \frac{1}{i}(\frac{\partial g}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial g}{\partial v}\frac{\partial v}{\partial y})) \\ &= \frac{1}{2}(\frac{\partial g}{\partial u}(\frac{\partial u}{\partial x} + \frac{1}{i}\frac{\partial u}{\partial y}) + \frac{\partial g}{\partial v}(\frac{\partial v}{\partial x} + \frac{1}{i}\frac{\partial v}{\partial y})) \\ &= \frac{\partial g}{\partial u}\frac{\partial u}{\partial z} + \frac{\partial g}{\partial v}\frac{\partial v}{\partial z} \end{aligned} \tag{1}$$

根据

$$\frac{\partial f}{\partial z} = \frac{\partial u}{\partial z} + i\frac{\partial v}{\partial z}, \frac{\partial \overline{f}}{\partial z} = \frac{\partial u}{\partial z} - i\frac{\partial v}{\partial z}$$

，可得

$$\frac{\partial u}{\partial z} = \frac{1}{2}(\frac{\partial f}{\partial z} + \frac{\partial \overline{f}}{\partial z}), \frac{\partial v}{\partial z} = -\frac{i}{2}(\frac{\partial f}{\partial z} - \frac{\partial \overline{f}}{\partial z}) \tag{2}$$

而根据

$$\frac{\partial g}{\partial z} = \frac{1}{2}(\frac{\partial g}{\partial u} + \frac{1}{i}\frac{\partial g}{\partial v}), \frac{\partial g}{\partial \overline{z}} = \frac{1}{2}(\frac{\partial g}{\partial u} - \frac{1}{i}\frac{\partial g}{\partial v})$$

有

$$\frac{\partial g}{\partial u} = \frac{\partial g}{\partial z} + \frac{\partial g}{\partial \overline{z}}, \frac{\partial g}{\partial v} = i(\frac{\partial g}{\partial z} - \frac{\partial g}{\partial \overline{z}}) \tag{3}$$

将 *(2)(3)* 代入 *(1)*，即有

$$\frac{\partial h}{\partial z} = \frac{\partial g}{\partial z}\frac{\partial f}{\partial z} + \frac{\partial g}{\partial \overline{z}}\frac{\partial \overline{f}}{\partial z}$$

第二个等式同理可得

9. Show that in polar coordinates, the Cauchy-Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{1}{r}\frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

Use these equations to show that the logarithm function defined by

$$\log z = \log r + i\theta \quad \text{where } z = re^{i\theta} \text{ with } -\pi < \theta < \pi$$

is holomorphic in the region $r > 0$ and $-\pi < \theta < \pi$.

证明 根据 $x = r\cos\theta, y = r\sin\theta$ 和链式法则，有

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cos\theta - \frac{\partial u}{\partial \theta}\frac{1}{r}\sin\theta, \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\sin\theta + \frac{\partial u}{\partial \theta}\frac{1}{r}\cos\theta$$

二者联立解得

$$\frac{\partial u}{\partial r}=\frac{\partial u}{\partial x}\cos\theta+\frac{\partial u}{\partial y}\sin\theta,\frac{1}{r}\frac{\partial u}{\partial\theta}=-\frac{\partial u}{\partial x}\sin\theta+\frac{\partial u}{\partial y}\cos\theta$$

类似的，可以得到

$$\frac{\partial v}{\partial r}=\frac{\partial v}{\partial x}\cos\theta+\frac{\partial v}{\partial y}\sin\theta,\frac{1}{r}\frac{\partial v}{\partial\theta}=-\frac{\partial v}{\partial x}\sin\theta+\frac{\partial v}{\partial y}\cos\theta$$

结合柯西-黎曼方程 $\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y},\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}$, 即有

$$\frac{\partial u}{\partial r}=\frac{1}{r}\frac{\partial v}{\partial\theta}\quad and\quad \frac{1}{r}\frac{\partial u}{\partial\theta}=-\frac{\partial v}{\partial r}.$$

对于 $\log z=\log r+i\theta$,$u=\log r,v=\theta$, 于是在定义域上 y,v 处处可微分，且满足

$$\frac{\partial u}{\partial r}=\frac{1}{r}\frac{\partial v}{\partial\theta}=\frac{1}{r}\quad and\quad \frac{1}{r}\frac{\partial u}{\partial\theta}=-\frac{\partial v}{\partial r}=0$$

于是 $\log z$ 是纯的

10. Show that

$$4\frac{\partial}{\partial z}\frac{\partial}{\partial\overline{z}}=4\frac{\partial}{\partial\overline{z}}\frac{\partial}{\partial z}=\triangle,$$

where $\triangle$ is the Laplacian

$$\triangle=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}.$$

证明 已知:

$$\frac{\partial}{\partial z}=\frac{1}{2}\left(\frac{\partial}{\partial x}-i\frac{\partial}{\partial y}\right),\quad \frac{\partial}{\partial\overline{z}}=\frac{1}{2}\left(\frac{\partial}{\partial x}+i\frac{\partial}{\partial y}\right).$$

于是

$$\begin{aligned}4\frac{\partial}{\partial z}\frac{\partial}{\partial\overline{z}}&=4\cdot\frac{1}{2}\left(\frac{\partial}{\partial x}-i\frac{\partial}{\partial y}\right)\cdot\frac{1}{2}\left(\frac{\partial}{\partial x}+i\frac{\partial}{\partial y}\right)\\&=\left(\frac{\partial}{\partial x}\right)^2-\left(i\frac{\partial}{\partial y}\right)^2\\&=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}=\triangle\end{aligned}$$

同理可证:

$$4\frac{\partial}{\partial\overline{z}}\frac{\partial}{\partial z}=\triangle.$$

16. Determine the radius of convergence of the series $\sum_{n=1}^{\infty} a_n z^n$ when:

(a) $a_n = (\log n)^2$

(b) $a_n = n!$

(c) $a_n = \frac{n^2}{4^n+3n}$

(d) $a_n = \frac{(n!)^3}{(3n)!}$ *[Hint: Use Stirling's formula, which says that* $n! \sim cn^{n+\frac{1}{2}}e^{-n}$ *for some* $c > 0$*.]*

(e) Find the radius of convergence of the hypergeometric series

$$F(\alpha,\beta,\gamma;z) = 1+\sum_{n=1}^{\infty} \frac{\alpha(\alpha+1)\cdots(\alpha+n-1)\beta(\beta+1)\cdots(\beta+n-1)}{n!\gamma(\gamma+1)\cdots(\gamma+n-1)} z^n.$$

Here $\alpha, \beta \in \mathbb{C}$ and $\gamma \neq 0, -1, -2, \ldots$

(f) Find the radius of convergence of the Bessel function of order r:

$$J_r(z) = \left(\frac{z}{2}\right)^r \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+r)!} \left(\frac{z}{2}\right)^{2n},$$

where r is a positive integer.

证明 *(a)*

$$|a_n|^{1/n} = \left((\log n)^{1/n}\right)^2$$

令 $t = \log n$，则 $n = e^t$，从而 $(\log n)^{1/n} = t^{1/e^t} = e^{\frac{\log t}{e^t}}$，当 $n \to \infty$ 时 $t \to \infty$，$\lim_{t\to\infty} \frac{\log t}{e^t} = 0$，故 $\lim_{n\to\infty}(\log n)^{1/n} = e^0 = 1$。因此：

$$\lim_{n\to\infty} |a_n|^{1/n} = 1^2 = 1$$

收敛半径：$R = \frac{1}{1} = 1$。

(b) 利用斯特林公式 $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$ ：

$$|a_n|^{1/n} = (n!)^{1/n} \sim \left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\right)^{1/n} = (2\pi n)^{1/(2n)} \cdot \frac{n}{e}$$

由于 $\lim_{n\to\infty}(2\pi n)^{1/(2n)} = 1$，故：

$$\lim_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty} \frac{n}{e} = \infty$$

因此：

$$R = 0$$

(c)

$$|a_n|^{1/n} = \left(\frac{n^2}{4^n + 3n}\right)^{1/n} = \frac{\left(n^{1/n}\right)^2}{\left(4^n\left(1 + \frac{3n}{4^n}\right)\right)^{1/n}} = \frac{\left(n^{1/n}\right)^2}{4 \cdot \left(1 + \frac{3n}{4^n}\right)^{1/n}}$$

而 $\lim_{n\to\infty} n^{1/n} = 1$，且 $\lim_{n\to\infty} \frac{3n}{4^n} = 0$，故 $\lim_{n\to\infty} \left(1 + \frac{3n}{4^n}\right)^{1/n} = 1$。因此：

$$\lim_{n\to\infty} |a_n|^{1/n} = \frac{1}{4 \cdot 1} = \frac{1}{4}$$

收敛半径：$R = \frac{1}{1/4} = 4$。

(d) 由斯特林公式：

$$(n!)^3 \sim (2\pi n)^{3/2}\left(\frac{n}{e}\right)^{3n}, \quad (3n)! \sim \sqrt{6\pi n}\left(\frac{3n}{e}\right)^{3n}$$

代入

$$|a_n|^{1/n} \sim \left(\frac{(2\pi n)^{3/2}\left(\frac{n}{e}\right)^{3n}}{\sqrt{6\pi n}\left(\frac{3n}{e}\right)^{3n}}\right)^{1/n} = \left(\frac{(2\pi)^{3/2}}{\sqrt{6\pi}}\right)^{1/n} n^{1/n} \cdot \frac{1}{27}$$

$\lim_{n\to\infty}(n)^{1/n} = 1$，故：

$$\lim_{n\to\infty} |a_n|^{1/n} = 1 \cdot \frac{1}{27} = \frac{1}{27}$$

收敛半径：$R = \frac{1}{1/27} = 27$。

(e) 超几何级数通项系数：

$$a_n = \frac{\alpha(\alpha+1)\cdots(\alpha+n-1)\beta(\beta+1)\cdots(\beta+n-1)}{n!\gamma(\gamma+1)\cdots(\gamma+n-1)}$$

若 $-\alpha \in \mathbb{N}$ 或 $-\beta \in \mathbb{N}$, 则从某一项起 $a_n = 0$, 从而收敛半径为 $R = \infty$

若 $-\alpha \notin \mathbb{N}$ 且 $-\beta \notin \mathbb{N}$ 时，有

$$\frac{a_{n+1}}{a_n} = \frac{(\alpha+n)(\beta+n)}{(n+1)(\gamma+n)} \to 1 \quad (n \to \infty)$$

因此：

$$\lim_{n\to\infty} |a_n|^{1/n} = 1$$

收敛半径：$R = 1$

(f)

$$J_r(z) = \left(\frac{z}{2}\right)^r \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+r)!} \left(\frac{z}{2}\right)^{2n} = \left(\frac{z}{2}\right)^r \sum_{n=0}^{\infty} a_n z^{2n}$$

其中 $a_n = \frac{(-1)^n}{n!(n+r)!4^n}$, 考虑级数 $\sum_{n=0}^{\infty} a_n w^n$

$$\frac{1}{R_w} = \limsup_{n\to\infty} |a_n|^{1/n} = \limsup_{n\to\infty} \left(\frac{1}{n!(n+r)!4^n}\right)^{1/n}.$$

由斯特林公式

$$\frac{1}{R_w} = \lim_{n\to\infty} \left(\frac{1}{n!(n+r)!4^n}\right)^{1/n} = \lim_{n\to\infty} \frac{e^2}{4n^2} = 0.$$

即有 $R_w = \infty$, 从而收敛半径 $R = \infty$

23. Consider the function f defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \le 0, \\ e^{-1/x^2} & \text{if } x > 0. \end{cases}$$

Prove that f is indefinitely differentiable on $\mathbb{R}$, and that $f^{(n)}(0) = 0$ for all $n \ge 1$. Conclude that f does not have a converging power series expansion $\sum_{n=0}^{\infty} a_n x^n$ for x near the origin.

证明 当 $x > 0$ 时, $f(x) = e^{-1/x^2}$ 是初等函数的复合, 显然无穷次可微, 且其各阶导数均为形如 $P(1/x)e^{-1/x^2}$ 的形式 ($P(t)$ 为关于 t 的多项式);

当 $x < 0$ 时, $f(x) = 0$, 显然无穷次可微, 且各阶导数均为 *0*;

当 $x = 0$ 时, $n = 1$ 时:

$$f'(0) = \lim_{x\to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0^+} \frac{e^{-1/x^2}}{x} = 0$$

假设对 $k \ge 1$, 有 $f^{(k)}(0) = 0$, 且当 $x > 0$ 时, $f^{(k)}(x) = P_k(1/x)e^{-1/x^2}$ ($P_k(t)$ 为多项式)。

$$f^{(k+1)}(0) = \lim_{x\to 0^+} \frac{f^{(k)}(x) - f^{(k)}(0)}{x - 0} = \lim_{x\to 0^+} \frac{P_k(1/x)e^{-1/x^2}}{x} = 0$$

由归纳法, 对所有 $n \ge 1$, $f^{(n)}(0) = 0$。

综上, *f is indefinitely differentiable on* $\mathbb{R}$

若 $f(x)$ 在原点附近有收敛的幂级数展开：

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n$$

由 $f^{(n)}(0)=0\ (n\geq 1)$，得 $f(x)=f(0)=0$，与 $x>0$ 时 $f(x)=e^{-1/x^2}>0$ 矛盾。

因此，$f(x)$ 在原点附近不存在收敛的幂级数展开。

25. The next three calculations provide some insight into Cauchy's theorem, which we treat in the next chapter.

(a) Evaluate the integrals

$$\int_\gamma z^n\,dz$$

for all integers n. Here γ is any circle centered at the origin with the positive (counterclockwise) orientation.

(b) Same question as before, but with γ any circle not containing the origin.

(c) Show that if $|a|<r<|b|$, then

$$\int_\gamma \frac{1}{(z-a)(z-b)}\,dz=\frac{2\pi i}{a-b},$$

where γ denotes the circle centered at the origin, of radius r, with the positive orientation.

证明 *(a)* 设 γ 的参数方程为 $z=re^{i\theta}\ (\theta\in[0,2\pi],\ r>0)$，则 $dz=ire^{i\theta}d\theta$，代入积分得：

$$\int_\gamma z^n\,dz=\int_0^{2\pi}(re^{i\theta})^n\cdot ire^{i\theta}d\theta=ir^{n+1}\int_0^{2\pi}e^{i(n+1)\theta}d\theta$$

当 $n\neq -1$ 时：

$$\int_0^{2\pi}e^{i(n+1)\theta}d\theta=\frac{1}{i(n+1)}\,e^{i(n+1)\theta}\Big|_0^{2\pi}=0$$

故 $\int_\gamma z^n\,dz=0$；

当 $n=-1$ 时：

$$\int_\gamma z^{-1}\,dz=ir^0\int_0^{2\pi}e^{i0\cdot\theta}d\theta=i\int_0^{2\pi}1\,d\theta=2\pi i$$

综上：

$$\int_\gamma z^n\,dz = \begin{cases} 2\pi i & n=-1, \\ 0 & n\in\mathbb{Z}, n\neq -1. \end{cases}$$

(b)

参数化 γ：$z(\theta)=z_0+Re^{i\theta}$，$\theta\in[0,2\pi]$，则 $dz=iRe^{i\theta}d\theta$。

积分化为：

$$\int_\gamma z^n\,dz = \int_0^{2\pi}(z_0+Re^{i\theta})^n\cdot iRe^{i\theta}d\theta$$

当 $n\neq -1$ 时：

$$\int_0^{2\pi}(z_0+Re^{i\theta})^n\cdot iRe^{i\theta}d\theta = \frac{1}{n+1}[(z_0+R)^{n+1}-(z_0+R)^{n+1}]=0$$

当 $n=-1$ 时：

$$\int_\gamma z^n\,dz = \int_0^{2\pi}\frac{iRe^{i\theta}}{z_0+Re^{i\theta}}d\theta = \ln(z_0+R)-\ln(z_0+R)=0$$

综上，对所有整数 n，均有：

$$\int_\gamma z^n\,dz = 0$$

(c)

$$\frac{1}{(z-a)(z-b)} = \frac{1}{a-b}\left(\frac{1}{z-a}-\frac{1}{z-b}\right)$$

$$\int_\gamma \frac{1}{z-a}\,dz = \int_0^{2\pi}\frac{1}{re^{i\theta}-a}\cdot ire^{i\theta}\,d\theta = i\int_0^{2\pi}\frac{re^{i\theta}}{re^{i\theta}-a}\,d\theta = i\int_0^{2\pi}\frac{1}{1-\frac{a}{r}e^{-i\theta}}\,d\theta.$$

由于 $|a|<r$，有 $\left|\frac{a}{r}e^{-i\theta}\right|<1$，可将被积函数展开为几何级数：

$$\frac{1}{1-\frac{a}{r}e^{-i\theta}} = \sum_{n=0}^{\infty}\left(\frac{a}{r}\right)^n e^{-in\theta},$$

逐项积分得：

$$i\sum_{n=0}^{\infty}\left(\frac{a}{r}\right)^n\int_0^{2\pi}e^{-in\theta}\,d\theta.$$

而

$$\int_0^{2\pi}e^{-in\theta}\,d\theta = \begin{cases} 2\pi, & n=0, \\ 0, & n\neq 0. \end{cases}$$

故

$$\int_\gamma \frac{1}{z-a}\,dz = 2\pi i.$$

由 *(b)* 的结论

$$\int_\gamma \frac{1}{z-b}dz = \int_{\gamma+b} \frac{1}{z}dz = 0$$

综上

$$\int_\gamma \frac{1}{(z-a)(z-b)}\,dz = \frac{1}{a-b}\left(\int_\gamma \frac{1}{z-a}dz - \int_\gamma \frac{1}{z-b}dz\right) = \frac{2\pi i}{a-b}$$